SANTOSH KUMAR KASAM

+91 9944533822 | kasamsantosh71@gmail.com | LinkedIn | Google Scholer

SUMMARY

AI Engineer with 4+ years of software engineering and AI experience, specializing in Enterprise Agentic AI, RAG architectures, and AI-driven workflow automation. Curious to learn new frameworks and build impacful solutions.

SKILLS

Proficient: LangChain, LangGraph, Copilot agent studio, PyTorch, Tensorflow, Streamlit, Python, C/C++.

Familier: Azure, Docker, Github actions, ZUUL, FastAPI, OpenCV, Letta AI, CrewAI.

WORK EXPERIENCE

Bertrandt Ingenieurbüro **Munich**

AI Engineer *Jun 2025 - Present*

- Designed and deployed a RAG agent that provides contextual project guidance and accelerates onboarding process for newly hired engineers.
- Architected and developed a RAG knowledge agent serving 40+ engineers, enabling semantic search and conversational access to project documentation, saving ~150 hours per month in search and knowledge exchange efforts.
- Built a Jira ticket handling agent that assists engineers with user story analysis, ticket refinement, and effort estimation, reducing impediments by ~30%, spillovers by ~60% and improving documentation quality.
- Led AI adoption initiatives as an AI Ambassador for a multinational project of 40+ engineers by identifying high-impact use cases, validating LLM-based solutions and mentoring engineers in AI-enhanced development workflows.
- Evaluated and integrated enterprise AI tools, including Microsoft Copilot and GitHub Copilot, and established best practices for secure and effective AI utilisation.
- Enforced Scrum principles as Scrum Master for a cross-functional and multinational feature team, improving delivery quality, Scrum sensitisation of the team members and identifying pain points through collaborative retros.

Function Developer *Aug 2022 - Jun 2025*

- Led the development and optimisation of the driving dynamics and height level functions, enhancing reliability and performance of ADAS systems for BMW's X3 and X5 models.
- Leveraged expertise in C/C++, unit and system testing, and code quality analysis to ensure robust software for safety approvals.
- Developed and integrated CAN and FlexRay communication networks for Driver Drowsiness Detection and Automatic Traction Control features within ASPICE-compliant development environments.
- Investigated and resolved communication-layer defects through structured root-cause analysis, improving robustness and maintainability of safety-critical automotive systems.
- Authored technical documentation and engineering guides that accelerated onboarding and reduced configuration-related errors for development teams.

Fraunhofer IMS **Duisburg**

Master Thesis - AI *Jan 2022 - Jul 2022*

- Developed an AI-based denoising pipeline for flash LiDAR point clouds, reducing false detections from 91% to 0.6%.
- Designed a multi-stage denoising algorithm that combines predictions from neural networks, adaptive thresholding, statistical filtering and 3D spatial analysis.
- Conducted extensive experimental evaluation on real-world sensor data to reduce overfitting and optimise accuracy.
- Published research outcomes in the IEEE Sensors Journal (2024): “MSNR: A Multistage Noise Reduction Method on Point Clouds”.

ZESS (Center for Sensor Systems) **Siegen**

Research Assistant - AI *Jun 2021 - Nov 2021*

- Developed and fine-tuned neural networks for pixel-wise material classification in Near-Infrared images using Transformer, CNN and ResNet architectures, achieving 95% classification accuracy.
- Built and curated large-scale datasets ranging from 100K to 1.7M samples, substantially improving model generalisation and pixel-level classification performance.
- Designed automated data-preparation and preprocessing pipelines, reducing dataset preparation effort by ~40 hours.
- Conducted comparative model evaluations based on accuracy, F1 score, computational complexity and deployment feasibility on GPUs.

EDUCATION

University of Siegen **Siegen, Germany**

M.Sc Mechatronics (Specialization in Deep Learning) *Graduation Date: Aug 2022*

Vellore Institute of Technology **Vellore, India**

B.Tech Mechanical Engineering *Graduation Date: May 2018*

PERSONAL STRENGTHS

- Clear and solution-oriented communication.
- High adaptability to changing project requirements.
- Ability to perform multiple roles simultaneously.